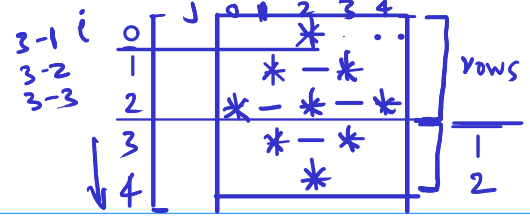
i-(rows-1)

2*rows-i+1

rows=3




```
for(int i=0; i<2*row-1 ; i++) {
    for(int j=0; j< 2*rows-1; j++) {
        if(i< rows){
            if(j < rows-i-1)
                printf(" ");
            else if( j < rows+i ) {
                if((i+j)%2==0)
                    printf("*");
                else
                    printf("-");
            }
        else{
            if(j<i-rows+1)
                printf(" ");
            else if( j<(2*rows-1)-(i-rows+1)) {
                if((i+j)%2==0)
                    printf("*");
                else
                    printf("-");
            }
        }
    }
    printf("\n");
}
```

rows

rows-1

first 3 rows

j <= rows+i-1

	0 1 2 3 4 5 6
0	*
1	* - *
2	* - * - *
3	* - * - * - *
4	* - * - *
5	* - *
6	*

rows=4

break, continue, goto, exit.

```
int i;
for( i=0 ; i<10; i++) {
    if(i>5)
        break;
    printf("%d", i);
}
printf("%d", i);
```

O/P

0 1 2 3 4 5 6

→ break — switch, or loop.

```
int i,j;
for(i=0; i<10; i++){
    for(j=0; j<10; j++){
        printf("%d", i+j);
        if(i+j>9)
            break;
    }
    printf("\n");
}
```

O/P

j	0	1	2	3	4	5	6	7	8	9
0	0	1	2	3	4	5	6	7	8	9
1	1	2	3	4	5	6	7	8	9	10
2	2	3	4	5	6	7	8	9	10	
3	3	4	5	6	7	8	9	10		
4	4	5	6	7	8	9	10			
5	5	6	7	8	9	10				
:	:									

```
for(int i=0; i<10; i++){
    if(i%2==0)
        continue;
    printf("%d", 2*i);
}
```

O/P

2 6 10 14 18

→ continues the next iteration
→ loop.

```
int i=0;
while(i<10){
    printf("%d", i);
    if(i%2==0)
        continue;
    i++;
}
```

O/P

0 0 0 0 0 0 0 0 0 0 0 0 0
0 0 0 0 0 0 0 0 0 0 0 0 0
0 0 0 0 0 . . . . . .

Infinite times

```
int i=0;
while(i<10){
    printf("%d", i);
    if(i%2==1)
        continue;
    i++;
}
```

O/P

0 1 1 1 1 1 1 1 1 1 1 1 1
1 1 1 1 1 . . . .

infinite times

```
int i=0;
while(i<10){
    i++;
    printf("%d", i);
    if(i%2==1)
        continue;
}
```

O/P

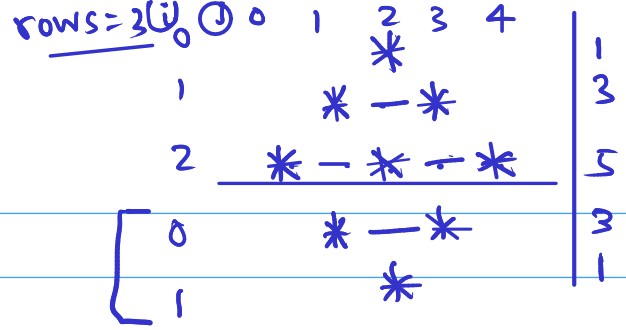
1 2 3 4 5 6 7 8 9 10

Useless

↓

Because no statements after continue in loop.

☆ one more way to do the same pattern




```
for(int i=0; i<rows; i++){
    for(int j=0; j<rows-i-1; j++)
        printf(" ");
    for(int k=0; k<2*i+1; k++){
        if(k%2)
            printf("-");
        else
            printf("*");
    }
    printf("\n");
}
for(int i=0; i<rows-1; i++){
    for(int j=0; j<i+1; j++)
        printf(" ");
    for(int k=0; k<(2*rows-1)-(2*(i+1)); k++){
        if(k%2)
            printf("-");
        else
            printf("*");
    }
    printf("\n");
}
```

$\frac{(2*rows-1)-2(1+i)}{}$

$5-2=3$

$5-4=1$